

ABSTRACT

This report presents the work carried out during a one-month Data Analyst internship at Advio Technologies, Hosur, where a complete sales data analytics pipeline was designed and implemented for a retail (Superstorestyle) transactional dataset. The objective of the internship project was to convert raw, unstructured sales records into actionable business insight using a combination of SQL, Python, and Microsoft Power BI, mirroring the analytics workflow followed by data teams in industry.

The dataset used for the analysis consisted of 150 transaction records spanning the calendar year 2025, covering three product categories (Furniture, Office Supplies and Technology), fifteen sub-categories, five regions and fifteen states across India. The work was carried out in four stages: (i) data cleaning and preprocessing, (ii) SQL-based aggregate querying to compute business metrics such as total sales, total profit, top customers and loss-making products, (iii) exploratory data analysis in Python using Pandas and Matplotlib to visualise monthly sales trends and state-wise profitability, and (iv) design of an interactive Power BI dashboard containing KPI cards, trend charts, category and regional breakdowns, and slicers for dynamic filtering.

The analysis revealed a total sales value of 59,13,245.24 and a total profit of ₹4,86,178.67, an overall profit margin of 8.22%. The Furniture category generated the highest sales (22,92,052.17) as well as the highest profit contribution, while the Consumer segment and the North region emerged as the strongest performing segment and region respectively. Ten customers were identified as high-value accounts contributing a disproportionately large share of total revenue, and state-wise profitability analysis highlighted Odisha, Maharashtra and Telangana as the most profitable states.

The internship provided practical, hands-on exposure to the complete data analytics lifecycle — from raw CSV data to a decision-ready interactive dashboard — and reinforced the importance of combining query-based aggregation, statistical/visual exploration, and business intelligence tools to support data-driven decision

1. INTRODUCTION

1.1 ABOUT THE ORGANIZATION

Adviyo Technologies is a software solutions firm based in Hosur, Tamil Nadu, offering web application development, mobile application development, and online marketing services to its clients. As part of its practical training programme, the organization provides student interns with hands-on exposure to real-world data handling and analytics tasks that mirror the requirements of business clients.

The one-month internship undertaken under the Data Analyst role was designed to give exposure to the complete lifecycle of a data analytics assignment — starting from a raw transactional dataset and ending with a decision-ready interactive dashboard, using the same tools (SQL, Python and Power BI) that are commonly used in the analytics industry.

1.2 INTERNSHIP OVERVIEW

During the internship, the intern was assigned a live-style project titled “Sales Performance Dashboard”, based on a retail transactional dataset. The assignment required the intern to independently clean the data, write SQL queries to extract key business metrics, perform exploratory data analysis in Python, and finally build an interactive Power BI dashboard summarising the findings for business stakeholders.

The internship followed an iterative, phase-wise structure: an initial phase for understanding the dataset and business requirements, a middle phase for SQL and Python-based analysis, and a final phase dedicated to dashboard design, formatting and presentation of insights.

1.3 PROBLEM STATEMENT

Retail businesses generate large volumes of transactional data across product categories, customer segments, and geographic regions on a continuous basis. Without a structured analytics process, it is difficult for decision-makers to quickly identify which categories, regions, or customers are driving profitability, which products or states are underperforming, and how sales vary across the year. The core problem addressed by this project was to

design a repeatable analytics pipeline that converts raw sales records into clear, quantified, and visual business insight that can support pricing, inventory, and marketing decisions.

1.4 OBJECTIVES OF THE PROJECT

- To clean and pre-process raw sales data to ensure accuracy and consistency for analysis.
- To write SQL queries that extract key performance metrics such as total sales, total profit, top customers, and profitability by state.
- To perform Exploratory Data Analysis (EDA) in Python to uncover seasonal and category-wise sales trends.
- To design and build an interactive Power BI dashboard containing KPI cards, trend charts, and slicers for stakeholders.
- To interpret the results and translate them into actionable business recommendations.

1.5 SCOPE OF THE PROJECT

The scope of the project covers the analysis of a retail transactional dataset containing 150 order records across three product categories, fifteen subcategories, five regions and fifteen states in India, for the calendar year 2025. The techniques demonstrated — SQL aggregation, Python-based EDA, and Power BI dashboard design — are generalisable and can be scaled to larger, real-time datasets in a production business intelligence environment.

2. LITERATURE REVIEW AND EXISTING SYSTEM

2.1 EXISTING SYSTEM

In many small and medium retail businesses, sales data is still analysed manually using spreadsheets. Reports are typically prepared periodically (weekly or monthly) by manually filtering and summarising rows in Excel. This existing approach has several limitations: it is time-consuming, errorprone, difficult to scale as data volume grows, and does not provide an interactive way for stakeholders to explore the data themselves. Manual spreadsheet-based reporting also makes it difficult to maintain a single, consistent version of the truth across different departments.

2.2 REVIEW OF DATA ANALYTICS PRACTICES

Modern data analytics practice addresses these limitations by separating the analytics workflow into distinct, well-defined stages: data storage and querying (using relational databases and SQL), statistical and exploratory analysis (typically using Python with libraries such as Pandas, NumPy and Matplotlib), and business intelligence visualisation (using tools such as Power BI, Tableau or Looker). SQL remains the standard language for aggregating and filtering large volumes of transactional data efficiently, while Python provides the flexibility needed for deeper statistical exploration and custom visualisation. Business intelligence dashboards, in turn, allow non-technical stakeholders to interact with the results directly through filters and drill-downs rather than depending on static reports.

This project follows the same layered approach that is widely used in the industry: SQL for aggregate metric extraction, Python for exploratory analysis, and Power BI for interactive presentation of insights to business users.

2.3 PROPOSED SYSTEM

The proposed system replaces manual, spreadsheet-based reporting with a structured three-stage analytics pipeline. Raw sales data is first queried using SQL to compute standard business metrics. The same data is then explored in Python to visualise trends that are not immediately obvious from tabular summaries, such as monthly seasonality and state-wise profit distribution. Finally, an interactive Power BI dashboard consolidates all key metrics and charts into a single, filterable interface, allowing stakeholders to explore sales performance by category, region, segment, and time period without needing to write queries themselves.

3. SYSTEM REQUIREMENTS AND TECHNOLOGY STACK

3.1 HARDWARE REQUIREMENTS

Component	Specification
Processor	Intel Core i3 (or equivalent) and above
RAM	8 GB minimum
Storage	256 GB (SSD recommended)
Operating System	Windows 10 / 11 (64-bit)

3.2 SOFTWARE REQUIREMENTS

Component	Specification
Programming Language	Python 3.x
Python Libraries	Pandas, NumPy, Matplotlib
Database / Query Language	SQL (MySQL / SQL Server compatible)
Business Intelligence Tool	Microsoft Power BI Desktop
Development Environment	Jupyter Notebook / VS Code
Data Format	CSV (Comma Separated Values)

3.3 TECHNOLOGY STACK OVERVIEW

The project uses a three-layer technology stack. The data layer consists of the raw sales dataset stored as a CSV file, structured in a single flat table ('superstore') suitable for direct SQL querying. The analysis layer uses SQL for aggregate business queries and Python (Pandas and Matplotlib) for exploratory data analysis and static chart generation. The presentation layer uses Microsoft Power BI Desktop to build an interactive dashboard with KPI cards, trend charts, and slicers, allowing business stakeholders to explore the data without technical expertise.

Pandas was chosen for data manipulation because of its efficient DataFrame-based operations for grouping, filtering, and aggregation, while Matplotlib was used for quick, static visualisations during the exploratory phase. Power BI was selected for the final dashboard because of its strong support for interactive slicers, drill-downs, and its wide adoption in industry as a standard business intelligence tool.

4. SYSTEM DESIGN AND METHODOLOGY

4.1 DATASET DESCRIPTION

The dataset used for this project, Superstore.csv, contains 150 retail transaction records for the calendar year 2025 (January to December), each representing one order line. It has 15 attributes describing the order, the customer, the product, and the financial outcome of the sale. Table 4.1 summarises the attributes used in the analysis.

Table 4.1: Dataset Attribute Description

Attribute	Type	Description
Order ID	Text	Unique identifier for each order
Order Date	Date	Date on which the order was placed
Ship Date	Date	Date on which the order was shipped
Customer Name	Text	Name of the customer
Segment	Text	Customer segment (Consumer, Corporate, Home Office)
Country / State / Region	Text	Geographic location of the order
Category	Text	Product category (Furniture, Office Supplies, Technology)
Sub-Category	Text	Product sub-category (e.g. Chairs, Phones, Binders)
Product Name	Text	Name of the product ordered
Sales	Numeric	Sales value of the order line, in ₹
Quantity	Numeric	Number of units ordered
Discount	Numeric	Discount fraction applied (0 to 1)
Profit	Numeric	Profit earned on the order line, in ₹

4.2 DATA ANALYTICS PIPELINE (METHODOLOGY)

The project followed a five-stage analytics pipeline, illustrated in Figure 4.1, moving from raw data collection through to actionable business insight:

- Stage 1 – Data Collection: The raw sales dataset was obtained in CSV format.

- Stage 2 – Data Cleaning: The dataset was checked for missing values, duplicate records, and incorrect data types, and the Order Date column was converted to a proper datetime format.
- Stage 3 – SQL Analysis: Aggregate SQL queries were used to compute business metrics such as total sales, total profit, and top customers directly on the structured data.
- Stage 4 – Python EDA: Pandas and Matplotlib were used to explore monthly sales trends and state-wise profitability visually.
- Stage 5 – Power BI Dashboard: All key metrics and charts were consolidated into a single interactive dashboard for business stakeholders.

Figure 4.1: Data Analytics Pipeline followed for the Project

Stage	Activity	Tool Used
1	Data Collection	CSV File
2	Data Cleaning & Preprocessing	Python (Pandas)
3	Aggregate Business Querying	SQL
4	Exploratory Data Analysis	Python (Pandas, Matplotlib)
5	Interactive Dashboard Design	Microsoft Power BI

4.3 DATA CLEANING AND PREPROCESSING

Before analysis, the dataset was inspected using `df.info()`, `df.describe()` and `df.isnull().sum()` in Pandas to check the structure, statistical summary, and presence of missing values respectively. The Order Date column, originally stored as text, was converted into a proper datetime object using `pd.to_datetime()` so that month-wise grouping could be performed accurately. No duplicate order records were found, and the numeric columns (Sales, Quantity, Discount, Profit) were verified to be of the correct data type before proceeding to the analysis stage.

4.4 DATABASE DESIGN

For the SQL analysis phase, the cleaned dataset was loaded into a single flat table named superstore, with each row representing one order line and each column corresponding to one of the fifteen dataset attributes described in Table 4.1. A single-table design was sufficient for this project since all analytical queries were aggregate queries (SUM, GROUP BY, ORDER BY)

operating on transactional data, without the need for joins across multiple normalised tables.

5. IMPLEMENTATION

5.1 SQL-BASED ANALYSIS

Eight aggregate SQL queries were written against the superstore table to extract the core business metrics required for the project — total sales, total profit, category-wise sales, top 10 customers, most profitable states, monthly sales, loss-making products, and region-wise sales. All queries use standard SUM(), GROUP BY, ORDER BY and LIMIT clauses. Representative queries are shown below, and the corresponding query outputs are shown in Figures 5.1 to 5.3.

```
SELECT SUM(Sales) AS Total_Sales FROM superstore;

SELECT SUM(Profit) AS Total_Profit FROM superstore;

SELECT Category, SUM(Sales) AS TotalSales
FROM superstore
GROUP BY Category
ORDER BY TotalSales DESC;

SELECT Customer_Name, SUM(Sales) AS TotalSales
FROM superstore
GROUP BY Customer_Name
ORDER BY TotalSales DESC
LIMIT 10;

SELECT State, SUM(Profit) AS Profit
FROM superstore
GROUP BY State
ORDER BY Profit DESC;

SELECT MONTH(Order_Date) AS Month, SUM(Sales) AS Sales
FROM superstore
GROUP BY Month;

SELECT Product_Name, SUM(Profit) AS Profit
FROM superstore
GROUP BY Product_Name
HAVING Profit < 0
ORDER BY Profit;

SELECT Region, SUM(Sales) AS Sales
FROM superstore
GROUP BY Region;
```

Region	Total_Sales	Total_Profit
Central	91506.79	16788.22
East	1601026.73	141152.85
North	1655558.19	126272.07
South	1381971.32	121680.67
West	1183182.21	80284.86

Figure 5.1: SQL Query Output – Region-wise Sales

Segment	Average_Discount
Consumer	0.162
Corporate	0.161
Home Office	0.135

Figure 5.2: SQL Query Output – Segment-wise Sales

Customer Name	Total_Orders
Nisha Roy	16
Pooja Nair	13
Deepak Y	11
Priya Sharma	10
Suresh B	9
Divya Iyer	9
Neha Gupta	9
Rahul Verma	8
Vikram Joshi	8
Manoj Kumar	7

Figure 5.3: SQL Query Output – Top 10 Customers

5.2 PYTHON-BASED EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was performed in Python using Pandas for data manipulation and Matplotlib for visualisation. After loading and inspecting the dataset, the total sales figure and category-wise sales totals were computed, followed by the identification of the top 10 customers by sales value. The Order Date column was converted to a datetime type to enable grouping by month, and a line chart of monthly sales was plotted to visualise seasonal trends. A bar chart of the top 10 states by profit was also generated to visualise state-wise profitability at a glance.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("sales_performance.csv")
df.info()
df.describe()
df.isnull().sum()

print(df['Sales'].sum())
category = df.groupby("Category")["Sales"].sum()

top = df.groupby("Customer Name")["Sales"].sum() \
      .sort_values(ascending=False).head(10)

df['Order Date'] = pd.to_datetime(df['Order Date'])
monthly = df.groupby(df['Order Date'].dt.month)['Sales'].sum()
monthly.plot(kind='line')
plt.title("Monthly Sales")
plt.show()

state = df.groupby("State")["Profit"].sum()
state.sort_values().tail(10).plot(kind='bar')
plt.show()
```

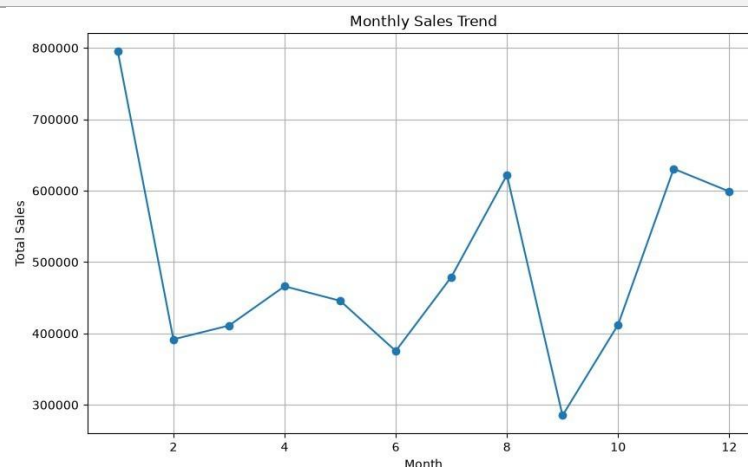


Figure 5.4: Python EDA – Monthly Sales Trend Line Chart

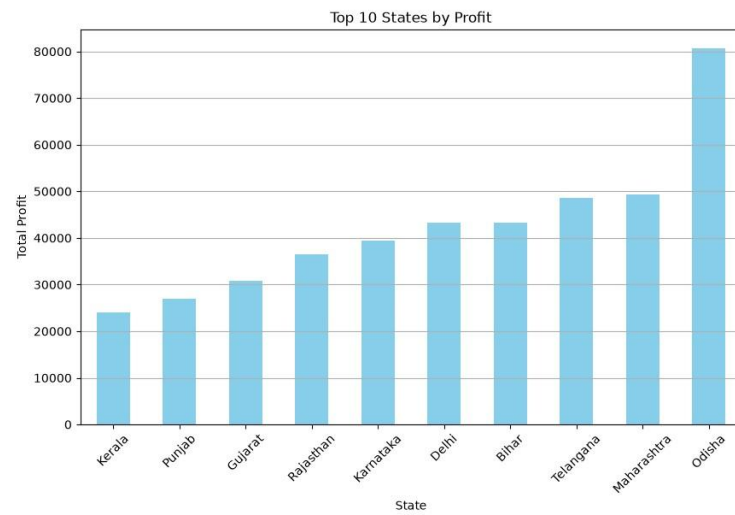


Figure 5.5: Python EDA – Top 10 States by Profit Bar Chart

5.3 POWER BI DASHBOARD DEVELOPMENT

The final stage of implementation involved building an interactive Power BI dashboard on top of the same dataset. Five KPI cards (Total Sales, Total Profit, Total Orders, Average Order Value, and Total Customers) were placed at the top of the dashboard, followed by a monthly sales trend line chart, a sales-by-category clustered column chart, a sales-by-region chart, a profit-by-state filled map, a top-10-customers bar chart, a sales-by-segment donut chart, a top-10-products bar chart, a loss-making-products table, and a discount-versus-profit scatter plot. Slicers for Order Date (Year), Region, Category, Segment, and State were added to allow stakeholders to interactively filter the entire dashboard. Figures 5.6 and 5.7 show the resulting dashboard views.

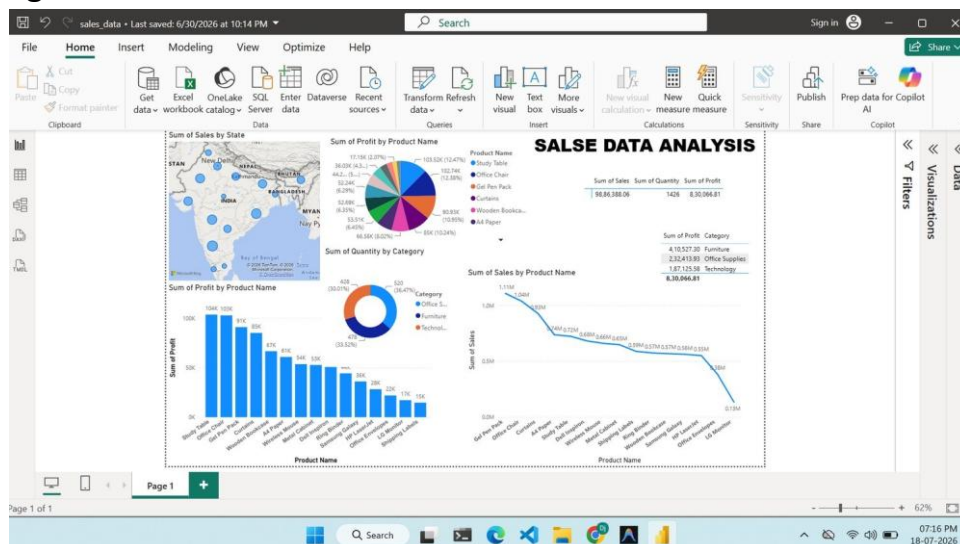


Figure 5.6: Power BI Dashboard – Sales Overview

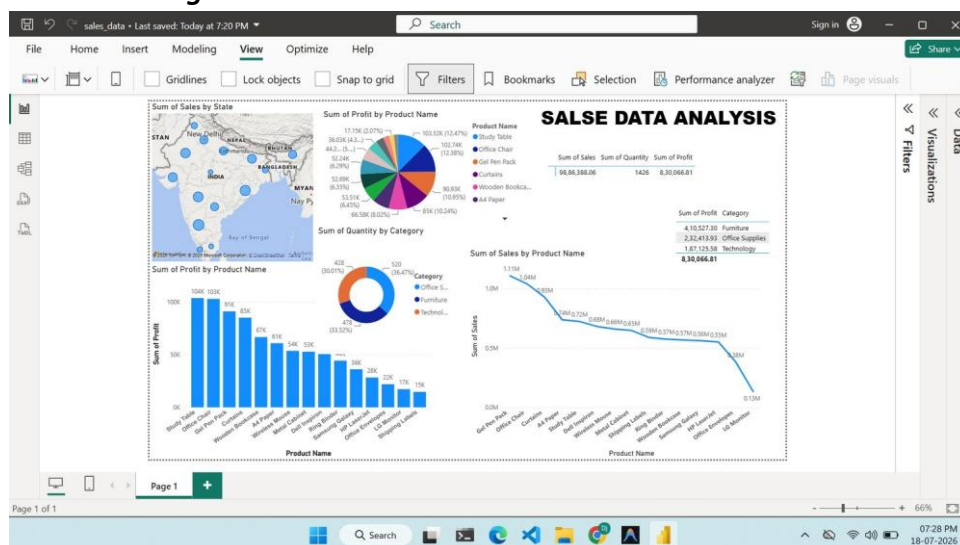


Figure 5.7: Power BI Dashboard – Regional and Profitability Insights

6. RESULTS AND DISCUSSION

6.1 KEY PERFORMANCE INDICATORS

Table 6.1 summarises the overall key performance indicators computed from the 150-record dataset for the year 2025.

Table 6.1: Key Performance Indicators Summary

Metric	Value
Total Sales	₹59,13,245.24
Total Profit	₹4,86,178.67
Overall Profit Margin	8.22%
Total Orders	150
Total Unique Customers	20
Average Order Value (AOV)	₹39,421.63

6.2 CATEGORY AND SUB-CATEGORY PERFORMANCE

Furniture was the highest-selling category as well as the highest profit contributor, followed by Office Supplies and Technology, as shown in Table 6.2. At the sub-category level (Table 6.3), Chairs and Tables generated the highest profit despite Pens having marginally higher sales than Chairs, indicating that Chairs and Tables carry a better margin within the Furniture category.

Table 6.2: Category-wise Sales and Profit ()₹

Category	Sales ()₹	Profit ()₹
Furniture	22,92,052.17	2,32,070.46
Office Supplies	19,49,252.68	1,34,391.16
Technology	16,71,940.39	1,19,717.05

Table 6.3: Sub-Category-wise Sales and Profit () – Top 10 by Sales₹

Sub-Category	Sales ()₹	Profit ()₹
Pens	6,29,393.51	51,068.71
Chairs	6,26,685.11	61,097.46

Sub-Category	Sales ()₹	Profit ()₹
Laptops	4,97,398.86	27,538.07
Furnishings	4,92,949.31	41,766.69
Storage	4,56,170.12	41,570.77
Paper	4,29,990.30	35,715.88
Accessories	3,95,902.11	35,104.71
Phones	3,91,256.24	27,913.32
Tables	3,87,935.04	53,594.50
Labels	3,29,298.54	11,295.98

6.3 REGION-WISE AND SEGMENT-WISE ANALYSIS

Table 6.4 shows that the North region contributed the highest sales, closely followed by the East region, while the Central region contributed noticeably less than the other four regions. Table 6.5 shows that the Consumer segment accounted for the largest share of sales, ahead of the Corporate and Home Office segments.

Table 6.4: Region-wise Sales Distribution ()₹

Region	Sales ()₹
North	16,55,558.19
East	16,01,026.73
South	13,81,971.32
West	11,83,182.21
Central	91,506.79

Table 6.5: Segment-wise Sales Distribution ()₹

Segment	Sales ()₹
Consumer	27,21,067.19
Corporate	16,29,005.94
Home Office	15,63,172.11

6.4 CUSTOMER ANALYSIS

Table 6.6 lists the top 10 customers ranked by total sales value. These ten customers represent high-value accounts and together account for a substantial share of total revenue, making them strong candidates for loyalty programmes or account-based marketing.

Table 6.6: Top 10 Customers by Sales Value ()₹

Customer Name	Sales ()₹
Nisha Roy	5,51,786.16
Pooja Nair	5,50,815.11
Suresh B	4,48,151.76
Priya Sharma	4,43,440.07
Deepak Y	3,88,506.87
Rahul Verma	3,69,305.21
Neha Gupta	3,67,817.04
Manoj Kumar	3,36,814.15
Vikram Joshi	3,32,106.35
Divya Iyer	2,88,866.40

6.5 STATE-WISE PROFITABILITY

Table 6.7 lists the ten most profitable states, led by Odisha, Maharashtra and Telangana. Table 6.8 lists the five states with the lowest (though still positive) profit contribution in this dataset, led by West Bengal and Tamil Nadu. Notably, no individual product in this dataset recorded a net loss across all its transactions, indicating that pricing and discounting were, overall, well controlled during the period analysed.

Table 6.7: Top 10 States by Profit ()₹

State	Profit ()₹
Odisha	80,674.41

Maharashtra	49,389.77
Telangana	48,676.00
Bihar	43,235.91
Delhi	43,191.79
State	Profit ()₹
Karnataka	39,415.39
Rajasthan	36,510.82
Gujarat	30,895.09
Punjab	26,949.80
Kerala	24,048.57

Table 6.8: Bottom 5 States by Profit ()₹

State	Profit ()₹
West Bengal	6,289.05
Tamil Nadu	9,540.71
Assam	10,953.48
Madhya Pradesh	16,788.22
Uttar Pradesh	19,619.66

6.6 MONTHLY SALES TREND

Table 6.9 and Figure 5.4 (Section 5.2) show that sales were highest in January, November and August, suggesting seasonal peaks possibly linked to year-start purchasing, pre-year-end promotions, and festive-season demand, while September recorded the lowest monthly sales in the dataset.

Table 6.9: Month-wise Sales Trend, 2025 ()₹

Month	Sales ()₹
January	7,94,854.96

February	3,91,779.71
March	4,11,050.08
April	4,66,101.64
May	4,45,788.76
June	3,75,430.48
July	4,79,131.51
August	6,22,051.46
September	2,85,135.16
October	4,12,326.40
Month	Sales ()₹
November	6,30,693.11
December	5,98,901.97

6.7 DISCOUNT VERSUS PROFIT RELATIONSHIP

The dataset showed an average discount rate of approximately 15.4% across all orders (ranging from 0% to 30%). A correlation analysis between Discount and Profit produced a coefficient of approximately -0.11, indicating only a weak negative relationship between the two variables in this dataset. This suggests that, within the discount range offered here, discounting alone was not a strong driver of reduced profitability, and other factors such as product category and cost structure played a larger role in determining profit outcomes.

6.8 DISCUSSION AND KEY INSIGHTS

- Furniture is both the top revenue-generating and top profit-generating category, making it a priority area for continued investment and inventory planning.

- The Consumer segment and North/East regions represent the strongest current markets and are good candidates for focused marketing campaigns.
- A small group of ten high-value customers contributes a disproportionately large share of revenue, supporting the case for a dedicated key-account or loyalty programme.
- Odisha, Maharashtra and Telangana stand out as the most profitable states and could be prioritised for expansion, while West Bengal and Tamil Nadu — though still profitable — may benefit from a review of local pricing and cost structures.
- January, August and November are peak sales months, useful for inventory and staffing planning ahead of these periods.
- Discount levels in this dataset show only a weak relationship with profit, suggesting the current discounting policy is not, by itself, a major risk to profitability.

7. CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

This internship project successfully implemented a complete, end-to-end sales data analytics pipeline covering data cleaning, SQL-based aggregate analysis, Python-based exploratory data analysis, and interactive Power BI dashboard design. Working with a 150-record retail transaction dataset for the year 2025, the project identified a total sales value of 59,13,245.24 and a total profit of ₹ ₹4,86,178.67, and surfaced clear, actionable insights on category performance, regional and segment-wise trends, high-value customers, statewide profitability, and seasonal sales patterns. The internship provided practical exposure to tools and techniques — SQL, Python, and Power BI — that are directly relevant to the role of a Data Analyst in industry, and reinforced the value of combining query-based, statistical, and visual analysis methods to support business decision-making.

7.2 LIMITATIONS

- The analysis is based on a relatively small dataset of 150 order records, which limits the statistical strength of trends such as monthly seasonality and the discount-profit relationship.

- The dataset covers a single calendar year (2025), so year-over-year growth trends could not be analysed.
- The Power BI dashboard, being a desktop file (.pbix), requires Power BI Desktop or Power BI Service to be viewed interactively and is not accessible as a live web dashboard in its current form.

7.3 FUTURE ENHANCEMENTS

- Scale the pipeline to a larger, real-time transactional dataset connected directly to a live production database.
- Automate the SQL-to-Python-to-Power BI workflow using a scheduled ETL (Extract, Transform, Load) process.
- Incorporate machine learning-based sales forecasting to predict future monthly sales and proactively plan inventory.
- Publish the Power BI dashboard to the Power BI Service to enable secure, web-based access for stakeholders.
- Add customer segmentation techniques such as RFM (Recency, Frequency, Monetary) analysis to further refine the high-value customer list.

APPENDIX 1: SQL QUERIES (ANALYSIS.SQL)

```
-- 1. Total Sales
SELECT SUM(Sales) AS Total_Sales FROM superstore;

-- 2. Total Profit
SELECT SUM(Profit) AS Total_Profit FROM superstore;

-- 3. Highest Sales Category
SELECT Category, SUM(Sales) AS TotalSales
FROM superstore GROUP BY Category ORDER BY TotalSales DESC;

-- 4. Top 10 Customers
SELECT Customer_Name, SUM(Sales) AS TotalSales
FROM superstore GROUP BY Customer_Name
ORDER BY TotalSales DESC LIMIT 10;

-- 5. Most Profitable State
SELECT State, SUM(Profit) AS Profit
FROM superstore GROUP BY State ORDER BY Profit DESC;

-- 6. Monthly Sales
SELECT MONTH(Order_Date) AS Month, SUM(Sales) AS Sales
FROM superstore GROUP BY Month;

-- 7. Loss-Making Products
SELECT Product_Name, SUM(Profit) AS Profit
FROM superstore GROUP BY Product_Name
HAVING Profit < 0 ORDER BY Profit;

-- 8. Region-wise Sales
SELECT Region, SUM(Sales) AS Sales
FROM superstore GROUP BY Region;
```

APPENDIX 2: PYTHON CODE (SALES_ANALYSIS.PY)

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("sales_performance.csv")
df.info()
df.describe()
df.head()
df.isnull().sum()

print(df['Sales'].sum())
category = df.groupby("Category")["Sales"].sum()
print(category)

top = df.groupby("Customer Name")["Sales"].sum() \
    .sort_values(ascending=False).head(10)
print(top)

df['Order Date'] = pd.to_datetime(df['Order Date'])
monthly = df.groupby(df['Order Date'].dt.month)['Sales'].sum()
monthly.plot(kind='line')
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()

state = df.groupby("State")["Profit"].sum()
state.sort_values().tail(10).plot(kind='bar')
plt.show()
```

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